Data Analytics (DA)

Module -1 - Excel

Excel Assignment 2 – Data Exploration

Tasks:-

Creating a Table Name as **Product_DB**

A	B	C	D	E
Product ID	Product Nam	Product Name_Standardization	Brand Nam	Price ($)
28-JAN-US	laptop	Laptop	Dell	1000
15-FEB-US	Sneakers	Sneakers	Nike	80
03-MAR-US	Coffee Maker	Coffee maker	Keurig	130
11-APR-US	smartphone	Smartphone	Samsung	900
22-MAY-US	Backpack	Backpack	North Face	70
07-JUN-UK	Headphones	Headphones	Sony	
19-JUL-UK	T-shirt	T-shirt	Adidas	30
23-AUG-UK	Blender	Blender	Ninja	90
05-SEP-UK	Tablet	Tablet	Apple	500
14-OCT-UK	Hiking Boots	Hiking boots	Timberland	130
17-JUN-IN	Laptop	Laptop	HP	950
25-NOV-AU	Sneakers	Sneakers	Adidas	90
08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120
18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150
16-APR-ES	headphones	Headphones	Bose	250
21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50
20-AUG-CN	Smartwatch	Smartwatch	Huawei	160
27-JAN-IT	laptop	Laptop	Asus	980
01-MAR-UK	Sunglasses	Sunglasses	Oakley	150

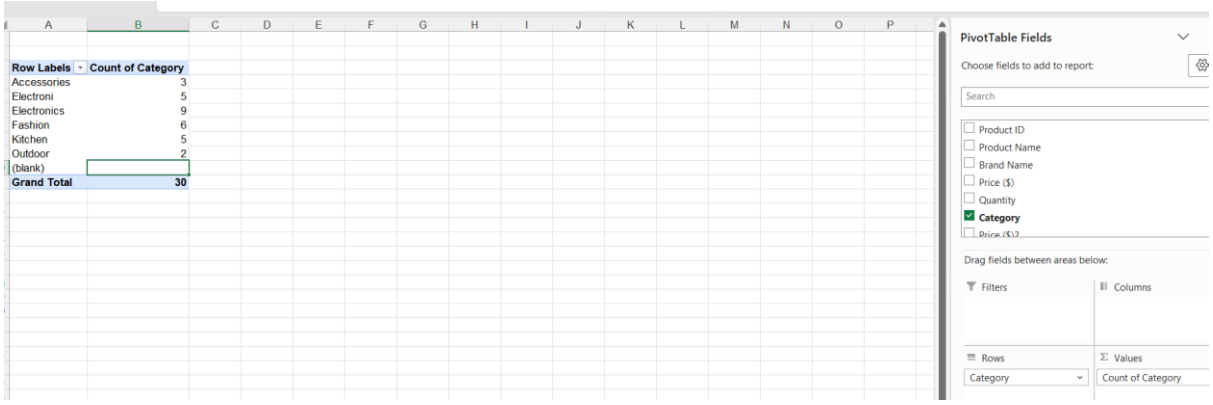
1. **Handling Missing Values:-**

● Checking for missing values in the 'Price' column. Using **Isblank** function to fill blank column and **Average** function to find the average price of the column.

F2 =ROUND(IF(ISBLANK([@[Price ($)]]),AVERAGE([Price ($)]),[@[Price ($)]]),0)

	A	B	C	D	E	F
1	Product ID	Product Nam	Product Name_Standardization	Brand Nam	Price ($)	Price_Clea
2	28-JAN-US	laptop	Laptop	Dell	1000	₹ 1,000.00
3	15-FEB-US	Sneakers	Sneakers	Nike	80	₹ 80.00
4	03-MAR-US	Coffee Maker	Coffee maker	Keurig	130	₹ 130.00
5	11-APR-US	smartphone	Smartphone	Samsung	900	₹ 900.00
6	22-MAY-US	Backpack	Backpack	North Face	70	₹ 70.00
7	07-JUN-UK	Headphones	Headphones	Sony		₹ 309.00
8	19-JUL-UK	T-shirt	T-shirt	Adidas	30	₹ 30.00
9	23-AUG-UK	Blender	Blender	Ninja	90	₹ 90.00
10	05-SEP-UK	Tablet	Tablet	Apple	500	₹ 500.00
11	14-OCT-UK	Hiking Boots	Hiking boots	Timberland	130	₹ 130.00
12	17-JUN-IN	Laptop	Laptop	HP	950	₹ 950.00
13	25-NOV-AU	Sneakers	Sneakers	Adidas	90	₹ 90.00
14	08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120	₹ 120.00
15	18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150	₹ 150.00
16	16-APR-ES	headphones	Headphones	Bose	250	₹ 250.00
17	21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50	₹ 50.00
18	20-AUG-CN	Smartwatch	Smartwatch	Huawei	160	₹ 160.00
19	27-JAN-IT	laptop	Laptop	Asus	980	₹ 980.00
20	01-MAR-UK	Sunglasses	Sunglasses	Oakley	150	₹ 150.00
21	14-AUG-US	Camping Tent	Camping tent	Coleman		₹ 309.00
22	14-MAY-RU	Camera	Camera	Nikon	700	₹ 700.00

● If there are products with missing categories, I used **Isblank** function to fill the column and used **Pivot table** to find the count of the category and I **choose top category** Electronics to fill the blanks.





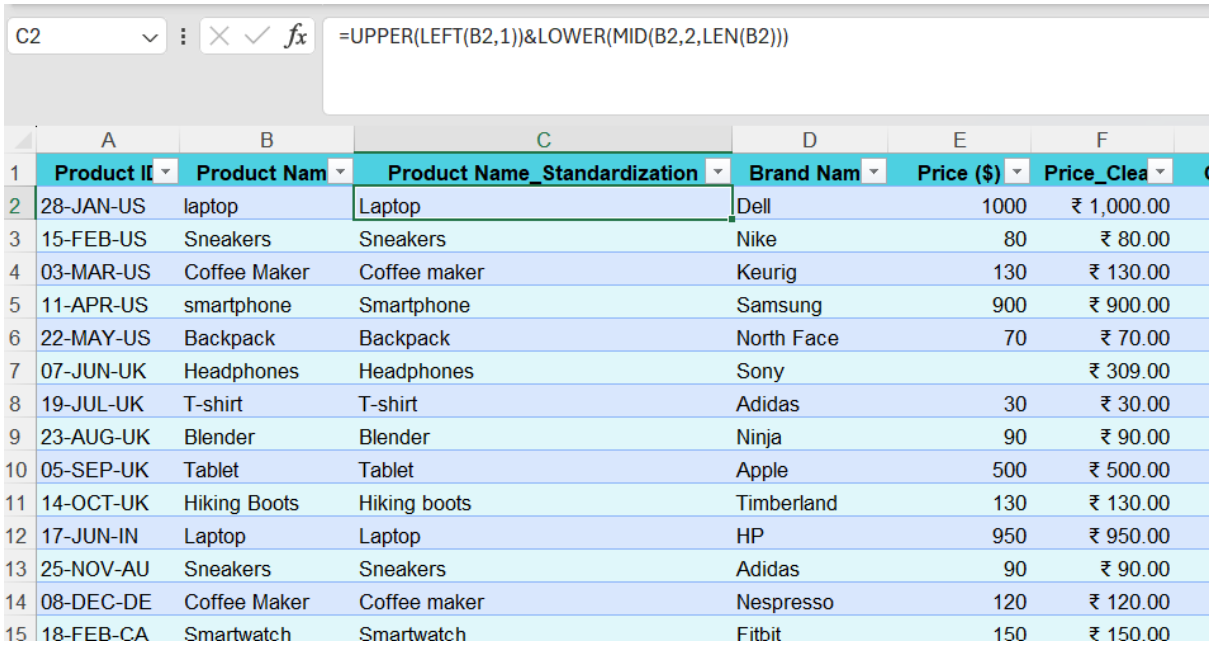
I2 =IF(ISBLANK([@Category]),"Electronics",[@Category])

	A	B	C	D	E	F	G	H	I
1	Product Id	Product Nam	Product Name_Standardization	Brand Nam	Price ($)	Price_Clea	Quantity	Category	Category_Clean
2	28-JAN-US	laptop	Laptop	Dell	1000	₹ 1,000.00	30	Electronics	Electronics
3	15-FEB-US	Sneakers	Sneakers	Nike	80	₹ 80.00	15	Fashion	Fashion
4	03-MAR-US	Coffee Maker	Coffee maker	Keurig	130	₹ 130.00	40	Kitchen	Kitchen
5	11-APR-US	smartphone	Smartphone	Samsung	900	₹ 900.00	25	Electronics	Electronics
6	22-MAY-US	Backpack	Backpack	North Face	70	₹ 70.00	20		Electronics
7	07-JUN-UK	Headphones	Headphones	Sony		₹ 309.00	45	Electroni	Electroni
8	19-JUL-UK	T-shirt	T-shirt	Adidas	30	₹ 30.00	5	Fashion	Fashion
9	23-AUG-UK	Blender	Blender	Ninja	90	₹ 90.00	35	Kitchen	Kitchen
10	05-SEP-UK	Tablet	Tablet	Apple	500	₹ 500.00	50	Electronics	Electronics
11	14-OCT-UK	Hiking Boots	Hiking boots	Timberland	130	₹ 130.00	10	Outdoor	Outdoor
12	17-JUN-IN	Laptop	Laptop	HP	950	₹ 950.00	25	Electronics	Electronics
13	25-NOV-AU	Sneakers	Sneakers	Adidas	90	₹ 90.00	40		Electronics
14	08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120	₹ 120.00	35		Electronics
15	18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150	₹ 150.00	15	Electronics	Electronics
16	16-APR-ES	headphones	Headphones	Bose	250	₹ 250.00	20	Electronics	Electronics
17	21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50	₹ 50.00	35	Accessories	Accessories
18	20-AUG-CN	Smartwatch	Smartwatch	Huawei	160	₹ 160.00	15	Electronics	Electronics
19	27-JAN-IT	laptop	Laptop	Asus	980	₹ 980.00	10	Electronics	Electronics

2. Correcting Inconsistent Data:-

● Identify any inconsistent text formats present in the "Product Name" column.

Created product name standardization column and using function Upper to change first letter in the product name in upper case.

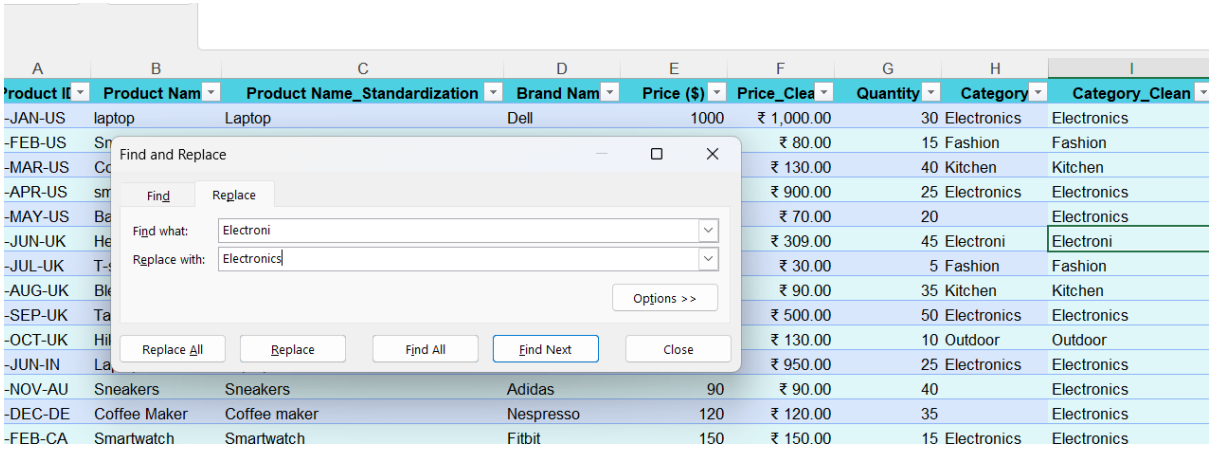


C2 =UPPER(LEFT(B2,1))&LOWER(MID(B2,2,LEN(B2)))

	A	B	C	D	E	F
1	Product ID	Product Nam	Product Name_Standardization	Brand Nam	Price ($)	Price_Clea
2	28-JAN-US	laptop	Laptop	Dell	1000	₹ 1,000.00
3	15-FEB-US	Sneakers	Sneakers	Nike	80	₹ 80.00
4	03-MAR-US	Coffee Maker	Coffee maker	Keurig	130	₹ 130.00
5	11-APR-US	smartphone	Smartphone	Samsung	900	₹ 900.00
6	22-MAY-US	Backpack	Backpack	North Face	70	₹ 70.00
7	07-JUN-UK	Headphones	Headphones	Sony		₹ 309.00
8	19-JUL-UK	T-shirt	T-shirt	Adidas	30	₹ 30.00
9	23-AUG-UK	Blender	Blender	Ninja	90	₹ 90.00
10	05-SEP-UK	Tablet	Tablet	Apple	500	₹ 500.00
11	14-OCT-UK	Hiking Boots	Hiking boots	Timberland	130	₹ 130.00
12	17-JUN-IN	Laptop	Laptop	HP	950	₹ 950.00
13	25-NOV-AU	Sneakers	Sneakers	Adidas	90	₹ 90.00
14	08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120	₹ 120.00
15	18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150	₹ 150.00

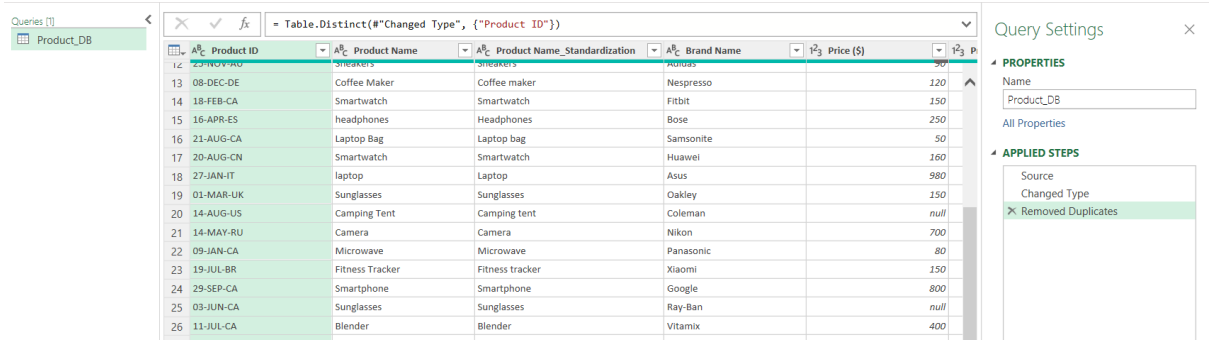
● Identify any typos present in the "Category" column.

Using find and replace to find the misspelling and replace the correct spelling ,change the Electroni into Electronics.



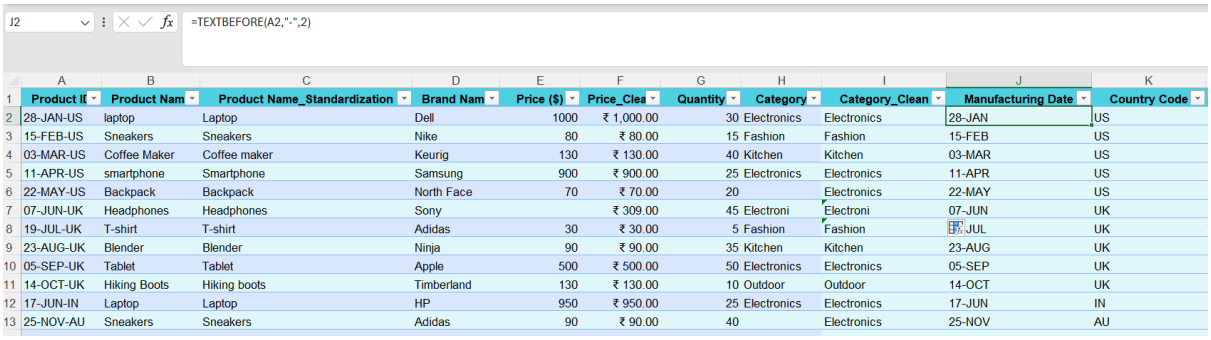
3. **Removing Duplicates:-**

● Identified the duplicate using power query and click the remove duplicates to remove the duplicate rows in the product id Column.



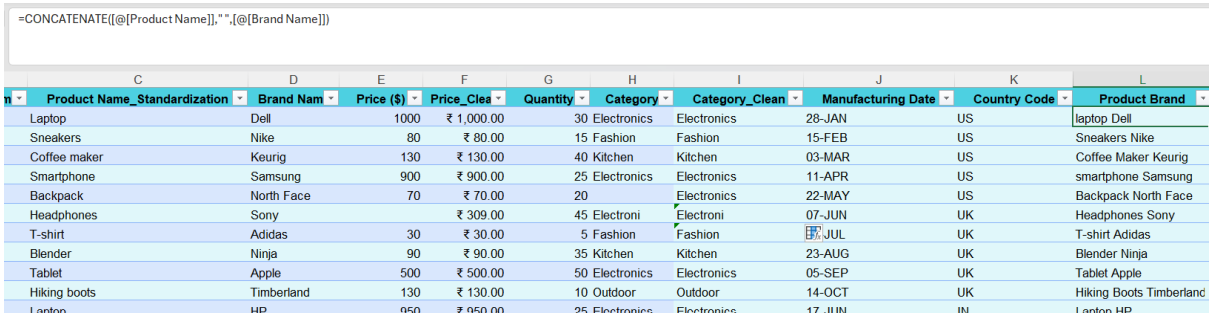
4. Splitting and Merging Data:-

● Spliting the "Product ID" column into two separate columns for " Manufacturing Date"
and "Country Code" using Textbefore and Textafter functions.



	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Nam	Product Name_Standardization	Brand Nam	Price ($)	Price_Clea	Quantity	Category	Category_Clean	Manufacturing Date	Country Code
2	28-JAN-US	laptop	Laptop	Dell	1000	₹ 1,000.00	30	Electronics	Electronics	28-JAN	US
3	15-FEB-US	Sneakers	Sneakers	Nike	80	₹ 80.00	15	Fashion	Fashion	15-FEB	US
4	03-MAR-US	Coffee Maker	Coffee maker	Keurig	130	₹ 130.00	40	Kitchen	Kitchen	03-MAR	US
5	11-APR-US	smartphone	Smartphone	Samsung	900	₹ 900.00	25	Electronics	Electronics	11-APR	US
6	22-MAY-US	Backpack	Backpack	North Face	70	₹ 70.00	20		Electronics	22-MAY	US
7	07-JUN-UK	Headphones	Headphones	Sony		₹ 309.00	45	Electroni	Electroni	07-JUN	UK
8	19-JUL-UK	T-shirt	T-shirt	Adidas	30	₹ 30.00	5	Fashion	Fashion	JUL	UK
9	23-AUG-UK	Blender	Blender	Ninja	90	₹ 90.00	35	Kitchen	Kitchen	23-AUG	UK
10	05-SEP-UK	Tablet	Tablet	Apple	500	₹ 500.00	50	Electronics	Electronics	05-SEP	UK
11	14-OCT-UK	Hiking Boots	Hiking boots	Timberland	130	₹ 130.00	10	Outdoor	Outdoor	14-OCT	UK
12	17-JUN-IN	Laptop	Laptop	HP	950	₹ 950.00	25	Electronics	Electronics	17-JUN	IN
13	25-NOV-AU	Sneakers	Sneakers	Adidas	90	₹ 90.00	40		Electronics	25-NOV	AU

● Merging the "Brand Name" and "Product Name" columns into one column named "Product Brand" using CONCATENATE function.



=CONCATENATE([@[Product Name]]," ",[@[Brand Name]])

	C	D	E	F	G	H	I	J	K	L
n	Product Name_Standardization	Brand Nam	Price ($)	Price_Clea	Quantity	Category	Category_Clean	Manufacturing Date	Country Code	Product Brand
	Laptop	Dell	1000	₹ 1,000.00	30	Electronics	Electronics	28-JAN	US	laptop Dell
	Sneakers	Nike	80	₹ 80.00	15	Fashion	Fashion	15-FEB	US	Sneakers Nike
	Coffee maker	Keurig	130	₹ 130.00	40	Kitchen	Kitchen	03-MAR	US	Coffee Maker Keurig
	Smartphone	Samsung	900	₹ 900.00	25	Electronics	Electronics	11-APR	US	smartphone Samsung
	Backpack	North Face	70	₹ 70.00	20		Electronics	22-MAY	US	Backpack North Face
	Headphones	Sony		₹ 309.00	45	Electroni	Electroni	07-JUN	UK	Headphones Sony
	T-shirt	Adidas	30	₹ 30.00	5	Fashion	Fashion	JUL	UK	T-shirt Adidas
	Blender	Ninja	90	₹ 90.00	35	Kitchen	Kitchen	23-AUG	UK	Blender Ninja
	Tablet	Apple	500	₹ 500.00	50	Electronics	Electronics	05-SEP	UK	Tablet Apple
	Hiking boots	Timberland	130	₹ 130.00	10	Outdoor	Outdoor	14-OCT	UK	Hiking Boots Timberland
	Laptop	HP	950	₹ 950.00	25	Electronics	Electronics	17-JUN	IN	Laptop HP

5. Number Formatting:-

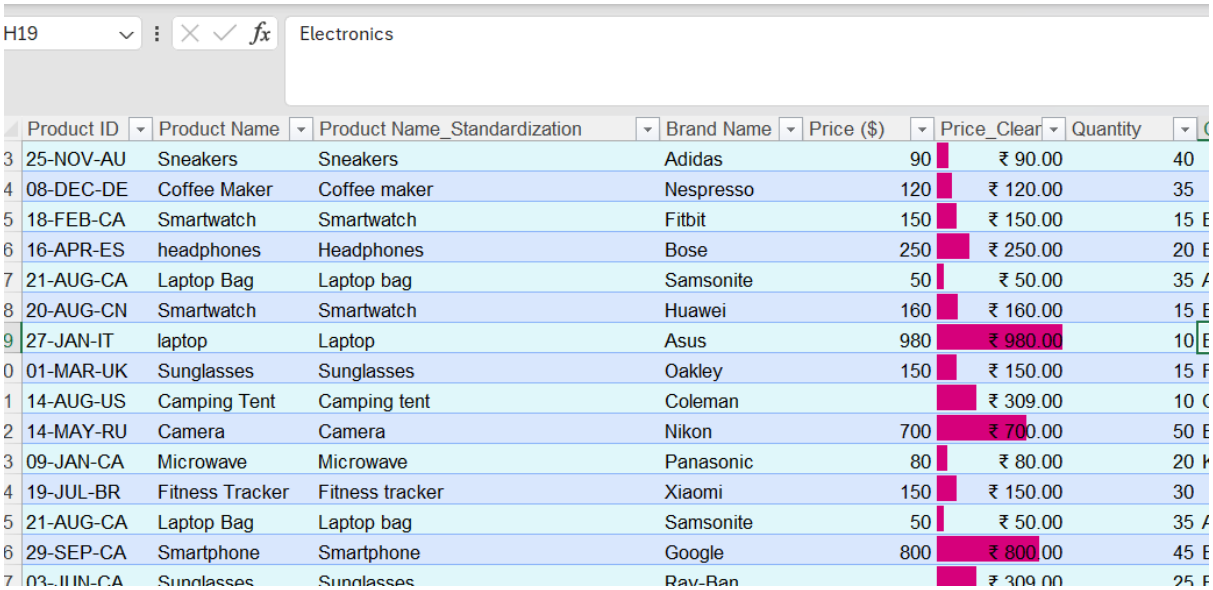
● Formating the data type of the "Price" column into a currency format first selecting the price column and click Home→General→Currency

	A	B	C	D	E	F
	Product ID	Product Nam	Product Name_Standardization	Brand Nam	Price ($)	Price_Clea
	28-JAN-US	laptop	Laptop	Dell	1000	₹ 1,000.00
	15-FEB-US	Sneakers	Sneakers	Nike	80	₹ 80.00
	03-MAR-US	Coffee Maker	Coffee maker	Keurig	130	₹ 130.00
	11-APR-US	smartphone	Smartphone	Samsung	900	₹ 900.00
	22-MAY-US	Backpack	Backpack	North Face	70	₹ 70.00
	07-JUN-UK	Headphones	Headphones	Sony		₹ 309.00
	19-JUL-UK	T-shirt	T-shirt	Adidas	30	₹ 30.00
	23-AUG-UK	Blender	Blender	Ninja	90	₹ 90.00
	05-SEP-UK	Tablet	Tablet	Apple	500	₹ 500.00
	14-OCT-UK	Hiking Boots	Hiking boots	Timberland	130	₹ 130.00
	17-JUN-IN	Laptop	Laptop	HP	950	₹ 950.00
	25-NOV-AU	Sneakers	Sneakers	Adidas	90	₹ 90.00
	08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120	₹ 120.00
	18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150	₹ 150.00

● Formating the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format. But Year is missing in the dataset

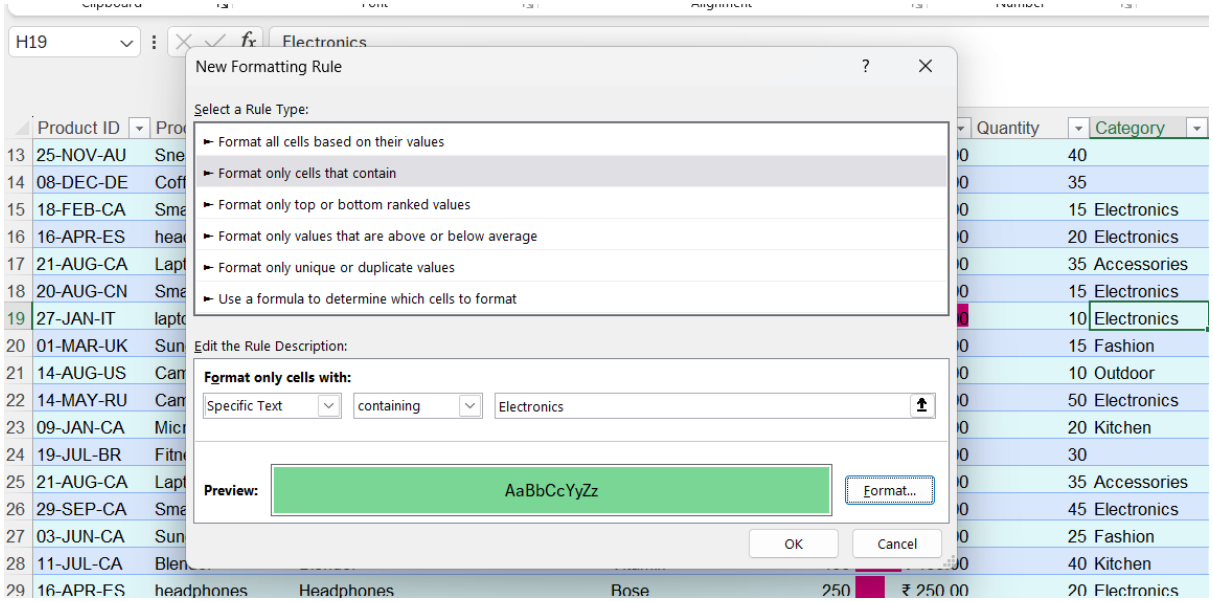
6. Conditional Formatting:-

● Applying data bar conditional formatting in the "Price" column.



	Product ID	Product Name	Product Name_Standardization	Brand Name	Price ($)	Price_Clear	Quantity
3	25-NOV-AU	Sneakers	Sneakers	Adidas	90	₹ 90.00	40
4	08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120	₹ 120.00	35
5	18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150	₹ 150.00	15
6	16-APR-ES	headphones	Headphones	Bose	250	₹ 250.00	20
7	21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50	₹ 50.00	35
8	20-AUG-CN	Smartwatch	Smartwatch	Huawei	160	₹ 160.00	15
9	27-JAN-IT	laptop	Laptop	Asus	980	₹ 980.00	10
0	01-MAR-UK	Sunglasses	Sunglasses	Oakley	150	₹ 150.00	15
1	14-AUG-US	Camping Tent	Camping tent	Coleman		₹ 309.00	10
2	14-MAY-RU	Camera	Camera	Nikon	700	₹ 700.00	50
3	09-JAN-CA	Microwave	Microwave	Panasonic	80	₹ 80.00	20
4	19-JUL-BR	Fitness Tracker	Fitness tracker	Xiaomi	150	₹ 150.00	30
5	21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50	₹ 50.00	35
6	29-SEP-CA	Smartphone	Smartphone	Google	800	₹ 800.00	45
7	03-JUN-CA	Sunglasses	Sunglasses	Ray-Ban		₹ 309.00	25

● Creating a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."



Thank you

J18 =TEXTBEFORE(A18,"-",2)

	Product ID	Product Name	Product Name_Standardization	Brand Name	Price ($)	Price_Clear	Quantity	Category	Category_Clean
13	25-NOV-AU	Sneakers	Sneakers	Adidas	90	₹ 90.00	40		Electronics
14	08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120	₹ 120.00	35		Electronics
15	18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150	₹ 150.00	15	Electronics	Electronics
16	16-APR-ES	headphones	Headphones	Bose	250	₹ 250.00	20	Electronics	Electronics
17	21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50	₹ 50.00	35	Accessories	Accessories
18	20-AUG-CN	Smartwatch	Smartwatch	Huawei	160	₹ 160.00	15	Electronics	Electronics
19	27-JAN-IT	laptop	Laptop	Asus	980	₹ 980.00	10	Electronics	Electronics
20	01-MAR-UK	Sunglasses	Sunglasses	Oakley	150	₹ 150.00	15	Fashion	Fashion
21	14-AUG-US	Camping Tent	Camping tent	Coleman		₹ 309.00	10	Outdoor	Outdoor
22	14-MAY-RU	Camera	Camera	Nikon	700	₹ 700.00	50	Electronics	Electronics
23	09-JAN-CA	Microwave	Microwave	Panasonic	80	₹ 80.00	20	Kitchen	Kitchen
24	19-JUL-BR	Fitness Tracker	Fitness tracker	Xiaomi	150	₹ 150.00	30		Electronics
25	21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50	₹ 50.00	35	Accessories	Accessories
26	29-SEP-CA	Smartphone	Smartphone	Google	800	₹ 800.00	45	Electronics	Electronics
27	03-JUN-CA	Sunglasses	Sunglasses	Ray-Ban		₹ 309.00	25	Fashion	Fashion
28	11-JUL-CA	Blender	Blender	Vitamix	400	₹ 400.00	40	Kitchen	Kitchen
29	16-APR-ES	headphones	Headphones	Bose	250	₹ 250.00	20	Electronics	Electronics